

HAMILTON ISLAND, Australia

WMO: 943680

Lat: 20.3662S

Lon: 148.9535E

Elev: 195

StdP: 14.59

Time Zone: 10.00 (E10)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	61.1	62.5	40.7	37.7	65.5	44.7	43.9	65.7	36.0	66.2	33.0	66.4	17.3	150	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	87.5	78.2	86.1	77.3	84.9	76.5	80.1	84.1	79.2	83.2	78.6	82.6	13.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	152.0	82.6	78.2	147.5	82.0	77.2	142.5	81.4	44.1	84.7	43.1	83.5	42.2	83.0	96.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
34.6	31.8	29.6	DB	59.4	92.0	2.2	4.2	57.9	95.1	56.6	97.5	55.3	99.9	53.7	102.9
			WB	48.6	83.5	2.3	5.2	47.0	87.2	45.6	90.2	44.3	93.1	42.7	96.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.2	81.0	80.8	79.4	76.6	72.5	68.8	67.4	68.7	72.5	75.8	78.2	80.6	(6)
	DBStd	5.35	2.03	2.09	2.26	1.99	2.52	2.71	2.49	2.51	2.38	2.04	1.99	2.02	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	14	0	0	0	0	0	3	8	3	0	0	0	0	(9)
	CDD50	9181	960	862	911	798	696	564	539	581	674	801	846	949	(10)
	CDD65	3721	495	442	446	348	232	117	82	119	224	336	396	484	(11)
	CDH74	24947	5001	4484	3915	1897	368	45	11	57	429	1441	2720	4579	(12)
	CDH80	4141	1114	989	575	74	2	0	0	0	10	99	318	960	(13)
(14) Wind	WSAvg	16.9	17.5	17.7	19.6	19.1	18.4	17.4	16.6	14.5	14.3	16.3	15.9	16.2	(14)
(15) Precipitation	PrecAvg	68.4	11.5	15.2	10.3	6.9	4.3	2.9	1.7	1.5	1.3	1.5	3.5	7.8	(15)
	PrecMax	127.6	26.5	37.3	39.2	15.8	14.3	9.3	7.2	6.7	10.4	6.9	28.8	52.4	(16)
	PrecMin	32.5	1.1	2.4	1.7	1.6	0.7	0.9	0.0	0.0	0.0	0.0	0.9	0.9	(17)
	PrecStd	23.5	7.3	10.0	8.9	3.6	2.7	2.0	1.7	1.7	2.0	1.7	5.6	9.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.2	88.2	87.6	83.7	80.2	77.3	75.6	77.6	81.0	84.4	86.4	89.2	(19)
		MCWB	79.3	78.8	79.8	75.0	72.7	70.7	68.4	68.5	70.7	73.2	74.3	77.6	(20)
	2%	DB	86.6	86.4	85.4	82.1	78.5	75.2	73.3	75.4	79.1	82.4	84.4	86.4	(21)
		MCWB	78.2	78.0	77.6	74.2	71.3	70.1	68.2	67.3	70.0	72.7	74.3	76.6	(22)
	5%	DB	85.9	85.8	84.2	80.7	77.1	73.6	71.8	73.7	77.8	80.8	82.7	85.7	(23)
		MCWB	77.6	77.7	76.6	73.2	70.7	69.4	66.9	66.6	69.3	72.1	73.2	76.1	(24)
	10%	DB	84.4	84.4	82.7	79.2	75.6	72.3	70.7	72.2	76.7	79.4	82.0	84.3	(25)
		MCWB	76.6	77.0	75.9	72.8	69.9	68.1	66.2	66.0	68.7	70.9	72.7	75.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	81.8	81.2	78.3	74.9	73.7	71.0	70.9	73.6	76.0	77.8	79.7	(27)
		MCDB	85.5	84.2	84.2	81.2	77.7	75.1	73.4	75.1	78.1	81.3	82.7	85.3	(28)
	2%	WB	80.1	80.1	79.1	76.5	73.5	71.9	69.8	69.4	72.4	74.6	76.4	78.6	(29)
		MCDB	84.4	83.8	82.5	78.9	76.4	73.7	71.8	73.1	76.7	79.4	81.7	83.6	(30)
	5%	WB	79.1	79.2	78.4	75.4	72.3	70.3	68.9	68.6	71.3	73.7	75.1	77.7	(31)
		MCDB	83.3	83.1	81.8	78.4	75.5	72.6	70.9	72.2	75.7	78.8	80.9	82.8	(32)
	10%	WB	78.3	78.5	77.6	74.5	71.4	69.2	67.7	67.4	70.2	72.5	74.1	76.9	(33)
		MCDB	82.6	82.5	81.2	77.9	74.7	71.7	69.8	71.0	74.6	77.9	80.0	82.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	6.9	6.1	5.6	5.4	5.1	5.2	6.6	7.7	7.9	7.7	7.9	(35)
		MCDBR	8.4	7.9	7.0	6.4	6.1	6.2	6.0	8.1	9.2	9.0	8.7	9.2	(36)
	5% WB	MCWBR	4.3	4.6	4.4	4.3	4.4	4.5	4.3	4.9	4.9	4.3	4.4	4.3	(37)
		MCWBR	7.8	7.4	6.8	5.7	5.6	5.5	5.3	7.3	8.3	8.4	8.7	8.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.399	0.397	0.388	0.361	0.345	0.333	0.320	0.327	0.355	0.380	0.385	0.395	(40)
		taud	2.542	2.549	2.560	2.620	2.624	2.648	2.663	2.633	2.536	2.492	2.518	2.522	(41)
	Ebn at Noon		300	297	292	288	281	279	287	295	297	298	302	301	(42)
		Edh at Noon	35	35	33	29	27	25	25	28	33	36	36	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2030	1920	1765	1574	1357	1216	1369	1635	1980	2188	2233	2228	(44)
	RadStd		168	213	174	142	90	119	98	108	121	120	255	233	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page